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Appendix 9.1	Representative Photographs of Fishponds within the Assessment Area

9. FISHERIES

9.1 Introduction

9.1.1.1 This section presents the fisheries baseline conditions within the Assessment Area, and the assessment results of the potential fisheries impacts resulting from the construction and operation of the Project. The location plan and description of the Project are presented in **Figure 1.1** and **Section 2**, respectively.

9.1.1.2 According to the EIAO-TM and Appendix H of the EIA Study Brief (ESB-373/2025), baseline conditions for fisheries resources were evaluated based on information from available literature and site visits conducted for the purposes of this EIA. Measures required to mitigate any identified adverse impacts are recommended, where appropriate, and residual impacts are assessed.

9.2 Environmental Legislation, Standards and Guidelines

9.2.1.1 Reference was made to the following HKSAR Government ordinances, regulations, standards, guidelines and documents when identifying, evaluating and assessing potential impacts of the Project on the fisheries resources:

- The Environmental Impact Assessment Ordinance (Cap. 499);
- Annexes 9 and 17 of the EIAO-TM;
- EIAO Guidance Note No. 15/2023 - Methodologies for Fisheries Baseline Surveys; and
- Water Pollution Control Ordinance (Cap. 358).

9.3 Assessment Methodology

9.3.1 Assessment Area

9.3.1.1 The Assessment Area includes areas within a 500 m distance from the proposed San Tin section of the NMH alignment and associated works (i.e. the Project Boundary) (**Figure 9**).

9.3.2 Methodology

9.3.2.1 Literature review and desktop study were conducted to collate the latest baseline conditions regarding culture fisheries resources within the Assessment Area. A list of available literature is provided in **Table 9.1** below.

Table 9.1 List of Relevant Literature

No.	Relevant Literatures
1.	Agriculture, Fisheries and Conservation Department (AFCD) Annual Reports 2015 – 2024 (AFCD, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024)
2.	Aquaculture Data from Agriculture, Fisheries and Conservation Department (AFCD) website (AFCD, 2026)
3.	Agriculture, Fisheries and Conversation Department (AFCD) Hong Kong Fisheries Profile 2024 (AFCD, 2025)
4.	EIA Report for Development at Ngau Tam Mei Area (Register No.: AEIAR-271/2025) (CEDD, 2025)

No.	Relevant Literatures
5.	EIA Report for Ngau Tam Mei Water Treatment Works Extension (Register No.: AEIAR-262/2024) (WSD, 2024)
6.	EIA Report for First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (Register No.: AEIAR-261/2024) (CEDD, 2024)
7.	EIA Report for Northern Link (Register No.: AEIAR-259/2024) (MTRCL, 2023)
8.	EIA Report of Housing Site in Yuen Long South (AEIAR-215/2017) (CEDD & PlanD, 2017)
9.	EIA Report for Proposed Low-rise and Low-density Residential Development at Various Lots and their Adjoining Government Land in D.D. 104, East of Kam Pok Road, Mai Po, Yuen Long. New Territories (Register No.: AEIAR-205/2017) (Glory Queen Limited, 2016)
10.	EIA Report for Proposed Residential Cum Passive Recreation Development within "Recreation" Zone and "Residential (Group C)" Zone at Various Lots in DD 104, Yuen Long, N.T. (Register No.: AEIAR-182/2014) (Capital Chance Ltd, 2013)
11.	EIA Report for North East New Territories New Development Areas (Register No.: AEIAR-175/2013) (CEDD, 2013)
12.	EIA Report of Hong Kong Section of Guangzhou-Shenzhen-Hong Kong Express Rail Link (AEIAR-143/2009) (MTRCL, 2009)
13.	EIA Report of Construction of Cycle Tracks and the Associated Supporting Facilities from Sha Po Tsuen to Shek Sheung River (Register No.: AEIAR-133/2009) (CEDD, 2008)
14.	EIA Report of Sheung Shui to Lok Ma Chau Spur Line (Register No.: AEIAR-052/2002) (KCRC, 2002)
15.	Recent aerial photos

9.3.2.2 In addition to the literature review with recent aerial photographs, fisheries baseline surveys in the form of site visits were also undertaken over a 9-month period, i.e. between April 2025 and December 2025, in accordance with the EIA Study Brief (ESB-373/2025) for the verification of the ponds and associated fisheries activities, with particular focus on those within the Project Boundary, where accessible. The conditions of the fishponds and the fisheries activities were observed and recorded. Drone photos were taken to facilitate the assessment of inaccessible ponds within the Assessment Area. Observation was made on the presence of any aquaculture practice and activities (e.g. draining, liming, re-profiling, stocking, feeding and harvesting), evidence of any recently used devices/equipment and sign of vegetation management.

9.3.2.3 General activity status of the fishponds was categorised as follows:

- **Active:** Utilised for commercial aquaculture activities, including commercial fishponds, fish fry ponds, and indoor fish farms;
- **Inactive:** No commercial aquaculture activities and management (e.g. overgrown vegetation at pond bunds hindering culture activities, and lack of equipment such as aerator). However, there should be no major physical constraints to its resumption in the short-term. Examples include fishponds with fish present in non-commercial quantities, and ponds for recreational fishing;
- **Abandoned:** Ponds with physical evidence that aquaculture has not been practised for years (typically ponds with overgrown vegetation or dried up) and/or obvious physical constraints to the resumption of fisheries activities (e.g. inaccessible or fenced off); and

- **Others:** Ponds unrelated to commercial aquaculture activities, including ponds created for wetland compensation or ecological enhancement, flood storage ponds, and wastewater treatment etc. These ponds, which were irrelevant to fisheries impacts, were scoped out from further assessment.

9.3.2.4 The status of fishpond management was investigated with reference made to the evidence of aquaculture activities, such as the presence of aquaculture operation (e.g. stocking, feeding, harvesting), management activities (e.g. pumping of water, draining, dredging, re-profiling and maintenance grass trimming), and the presence of any associated equipment (e.g. aerators, conditions of pond bund and fodder) and their conditions. Attention was also given to any watercourses which potentially serve as water sources for fishponds.

9.4 Fisheries Baseline Condition from Literature Review

9.4.1 Description of the Environment and Baseline Condition

Aquaculture (Pond Fish Culture) Activities and Resources

9.4.1.1 Fishponds were mainly identified at the southeast of Nam Sang Wai (NSW), west of the existing San Tin Highway, and west of the Ngau Tam Mei Water Treatment Works (NTMWTW). Some scattered fishponds were recorded at the northeast of the San Tin Barracks, west of the Sheung Yue River (SYR), and Tit Hang. Pond fish culture of various statuses was identified, including active, inactive and abandoned fishponds (**Figures 9.1** to **9.9** refer).

Other Fisheries Resources

9.4.1.2 No other fisheries resources such as capture fisheries or other aquaculture (e.g. Fish Culture Zones or oyster culture) were recorded within the Assessment Area. Considering the absence of these other fisheries resources, potential impacts on these resources were not anticipated. Therefore, no further detailed assessment was required in these other fisheries resources (**Figures 9.1** to **9.9** refer).

9.4.2 Pond Culture Fisheries Activities and Resources

Desktop Study

9.4.2.1 Pond fish culture in Hong Kong has been centred in the northwest New Territories. About 88% of the fish farms are engaged in polyculture (including bighead carp, grass carp, common carp and silver carp in combination with tilapia or grey mullet). The remaining 12% practise monoculture of carnivorous species such as giant groupers, seabreams and spotted scat in brackish fishponds near the coastline (AFCD, 2026).

9.4.2.2 According to the data from AFCD on aquaculture in Hong Kong, the production of pond fish was stabilised until 2021 but has declined significantly since 2022. Annual pond fish production and fishpond areas in the territory are listed in **Table 9.2** below. Local pond fish production only accounted for about 3.9% of total consumption of freshwater fish in 2024 (AFCD, 2025).

Table 9.2 Annual Pond Fish Production and Fishpond Areas between 2015 and 2025

Year	Pond Fish Production (tonne)	Fishpond Area (ha)
2025	1,518	1,115
2024	1,349	1,119
2023	1,050	1,130
2022	2,100	1,130
2021	2,900	1,130
2020	2,500	1,130
2019	2,300	1,130
2018	2,500	1,130
2017	2,543	1,132
2016	2,543	1,135

Source: AFCD, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026

9.5 Fisheries Baseline Condition from Recent Survey

9.5.1 Recent Site Visit Findings

9.5.1.1 The distribution and physical conditions of the fishponds are largely similar and comparable to the baseline conditions identified from literature review, which are considered relevant for the assessment. Based on observations from site visits, fishponds (including active, inactive and abandoned fishponds) and other non-fisheries related ponds were recorded across the Assessment Area, numbered as P#1 – P#258 (**Figures 9.1** to **9.9** refer).

9.5.1.2 Among the fishponds recorded within the Project Boundary, majority of the fishponds were active fishponds. Active fishponds identified within the Project Boundary were concentrated to the west of NTMWTW and some scattered ones west of SYR. Direct rainfall was the major water source of fishponds within the Assessment Area. Other potential water sources for fishponds include wells and from adjacent fishponds where water was pumped and drained from one pond to another.

9.5.1.3 Apart from fishponds, ten ponds were identified as “Other Ponds (non-fisheries related)”, including P#48 and P#49 (non-fisheries related ponds within Park Yoho and Sha Po Marsh area, **Figure 9.2**), P#58, P#61-#64 (Yuen Long Bypass Floodway Engineered Wetland, **Figure 9.2**), P#84 and P#85 (Pok Wai Floodwater Pumping Station, **Figure 9.3**), and P#247 (i.e. Kwu Tung Reservoir, **Figure 9.8**). Except P#247, all of these “Other Ponds (non-fisheries related)” were recorded outside the Project Boundary.

9.5.1.4 A total of 258 ponds were identified within the Assessment Area. Among these ponds, 34 ponds (including 28 active fishponds, four inactive fishponds, one abandoned fishpond, one “other pond”) fall wholly or partially within the Project Boundary. Among the 28 active fishponds, five of them are excluded from land resumption and no direct impact is expected to these fishponds. The active fishponds accounts for 36.3% in terms of pond areas within Assessment Area, and 44.2% within the Project Boundary (**Table 9.3**).

9.5.1.5 Within the Assessment Area, fisheries activities were observed concentrated in Ngau Tam Mei (NTM) area and to the west of San Tin Highway. In addition, fisheries activities were also observed scattered in area across the Assessment Area such as Ki Lun Tsuen, Tung Shing Lei, AFCD Au Tau Fisheries Office, to the west of SYR and Tit Hang.

9.5.2 Summary of Fisheries Resources

9.5.2.1 The locations and statuses of the fishponds and other ponds within the Assessment Area are indicated in **Figures 9.1** to **9.9**. Representative photos showing different types of fishponds are shown in **Appendix 9.1**. Details of the size and number of ponds recorded are presented in **Table 9.3** below.

Table 9.3 Summary of Ponds Identified within the Assessment Area and Project Boundary

Pond Status	Within Assessment Area		Within Project Boundary (1)	
	Size of Pond (2) (ha) (%)	Number of Pond	Size of Pond (2)(3) (ha) (%)	Number of Pond
Active Fishponds	16.7 (36.3%)	133	1.9 (44.2%)	28
Inactive Fishponds	13.5 (29.3%)	56	0.2 (4.7%)	4
Abandoned Fishponds	8.4 (18.3%)	59	0.7 (16.3%)	1
Other Ponds (Non-fisheries related)	7.4 (16.1%)	10	1.5 (34.9%)	1
Total	**46.0 (100%)**	**258**	**4.3 (100%)**	**34**

Notes:
(1) Areas of ponds within Project Boundary that overlap with site areas of concurrent project, i.e. Development at Ngau Tam Mei Area, are excluded in the calculation as they would be lost under the concurrent project.
(2) The size and percentage included in this table have been subject to rounding adjustments. Any discrepancies between total and sums of individual numbers listed therein are due to rounding.
(3) Loss of whole pond has been assumed for partial encroachment of Project Boundary onto pond.

Active Ponds

9.5.2.2 Active fishponds were predominantly concentrated in Ngau Tam Mei (NTM) area and under active production. Some active pond in NTM and at the west of the NTMWTW fall within the Project Boundary (i.e. P#163-168 and P#186-200). Other active fishponds in NTM area were located outside the Project Boundary (i.e. P#111-122, P#125-128, P#130, P#137-140, P#143, P#146-153, P#176-178, P#201, P#208-211 and P#250-257) (**Figure 9.4** refers). Apart from NTM area, active fishponds were also recorded in Ki Lun Tsuen, within the Project Boundary (i.e. P#242-243). In some other areas, scattered active fishponds under active production were also identified outside the Project Boundary, including Tung Shing Lei (i.e. P#17-20), AFCD Au Tau Fisheries Office (i.e. P#27-42), southeast of NSW (i.e. P#60 and P#65), to the west of San Tin Highway (i.e. P#71-76, P#79-83, P#89, P#106-107 and P#249), to the west of SYR (i.e. P#234-239 and P#241) and Tit Hang (i.e. P#248). Active fishponds within the Assessment Area were found to be utilised for cultivation of edible fish species, such as Bighead Carp, Jade Perch and Grass Carp. Quite a number of active fishponds in NTM area, including active

fishponds within Project Boundary (e.g. P#154-157, P#163-168, P#170-173, P#186-190, P#191-194 and P#195-200), were recorded to cultivate non-edible ornamental fish species, such as Koi.

9.5.2.3 To minimise the direct impact to active fishponds and the corresponding fisheries production, P#170-173 and P#244 are excluded from land resumption and no direct impact to these five ponds is expected. On the other hand, some of the active fishponds (i.e. P#108-110, P#141-142, P#154-158, P#175, P#184 and P#212-213) were within the site areas of concurrent project in NTM area (i.e. Development at Ngau Tam Mei Area) where the fishponds are expected to be lost under the concurrent project.

Inactive Ponds

9.5.2.4 Inactive fishponds were scattered across the Assessment Area, including those along Kam Tin Main Drainage Channel (KTMDC) and near NSW (P#2-4, P#26, P#44, P#50-53, P#55-57, P#59 and P#67-70), and the west of San Tin Highway near Fairview Park Boulevard (i.e. P#87-88, P#90-94, P#103 and P#105). A few were scattered in NTM area (i.e. P#129, P#131-133, P#160-162, P#169, P#174, P#179-183, P#203 and P#205-207), while some were near the northeast of the San Tin Barracks (i.e. P#216-220, P#227 and P#229) and west of the SYR (i.e. P#232-233, P#240 and P#246) (**Figure 9.4** refers). Except P#160-162 and P#169 which fall within the Project Boundary to the west of the NTMWTW, other inactive fishponds were located outside the Project Boundary. One inactive fishpond (i.e. P#159) was within the site areas of concurrent project in NTM area (i.e. Development at Ngau Tam Mei Area) where the fishpond is expected to be lost under the concurrent project..

9.5.2.5 These inactive fishponds were observed with no current utilisation for commercial aquaculture activities and lack of management for pond fish culture, but the pond structure remained relatively intact, with potential to be resumed for fisheries production with maintenance and management.

Abandoned Ponds

9.5.2.6 Abandoned fishponds were also recorded across the Assessment Area, including those along KTMDC (i.e. P#1, P#5-16, P#21-25, P#43 and P#45-47), to the southeast of NSW (i.e. P#54 and P#66), west of San Tin Highway (i.e. P#77-78, P#86, P#95-102 and P#104), in NTM area (i.e. P#123-124, P#134-136, P#185, P#202 and P#204) and northeast of San Tin Barracks (i.e. P#214-215, P#221-226, P#228, P#230-231, P#245 and P#258) (**Figure 9.4** refers). Except P#185 which fall within the Project Boundary identified at the west of the NTMWTW, other abandoned ponds were located outside the Project Boundary. Two abandoned fishponds were within the site areas of concurrent project in NTM area (i.e. P#144-145) where the fishponds are expected to be lost under the concurrent project.

9.5.2.7 All of these abandoned fishponds were observed with no current utilisation for commercial aquaculture activities and with overgrown vegetation. The abandoned and overgrown structure of these fishponds would be a physical constraint to the resumption of fisheries activities in a short period of time.

9.6 Identification and Evaluation of Potential Impacts

9.6.1 Identification of Potential Impacts

9.6.1.1 Given that there are no capture fisheries or other aquaculture such as Fish Culture Zones, oyster culture, sites of fisheries importance within the Assessment Area, direct and indirect impacts to these resources are not anticipated. Potential impacts arising from the Project would be limited to the loss of fish culture area (i.e. fishponds), aquaculture activities and aquaculture potential within the Project Boundary due to land resumption. The direct impacts and other associated indirect impacts potentially occurring during the construction and operational phases are listed in **Table 9.4** below.

Table 9.4 Potential Impacts during the Construction and Operational Phase

Phase	Direct Impact	Indirect Impact
Construction	• Loss of active fishponds and associated fisheries activities and production • Loss of inactive fishponds • Loss of abandoned fishponds • Loss of watercourses for aquaculture activities	• Deterioration of water quality • Potential water table drawdown • Potential vibration impact • Bund stability • Blockage of access
Operational	Nil	Nil

9.6.1.2 Details of each identified potential impact on fisheries resources are further described and evaluated in the following sections, with reference made to the evaluation criteria denoted in the EIAO-TM. A summary of impacts on fisheries resources is provided in **Table 9.6** to **Table 9.8**.

9.6.2 Construction Phase – Direct Impact

9.6.2.1 Under the Project, fishponds within the Project Boundary would be subjected to land resumption and would no longer be available for aquaculture activities, thus, would be regarded as direct loss of fishponds and associated fisheries activities, resources and production, and direct loss of aquaculture potential. Despite the proposed alignment would not be encroached into the fishponds to be impacted by temporary works, these fishponds are proposed to be included in the proposed wetland compensation area after the completion of construction works as an ecological enhancement measure of the overall NMH Project as detailed under **Section 8.11** of this report. Key fisheries activities, resources and production within the Project Boundary, including active fishponds and inactive fishponds, were mainly situated in NTM area while a few were located in Ki Lun Tsuen area.

Table 9.5 Direct Loss of Fishponds **within the Project Boundary**

Pond Status	Direct Loss of Fishponds from Proposed Alignment (i.e. at-grade works (i.e. roads, slope works, tunnel portals, ancillary facilities, etc) and viaduct supports (i.e. pile and pier related works))		Direct Loss of Fishponds from Temporary Works (i.e. works area, access and storage area) [1]		Overall	
	Size of Pond [2][3](ha) (%)	Number of Pond	Size of Pond [2][3](ha) (%)	Number of Pond	Size of Pond [2][3](ha) (%)	Number of Pond
Active Fishponds	0.9 (52.9%)	10	0.7 (87.5%)	13	**1.5 (62.5%)**	**23**
Inactive Fishponds	0.1 (5.9%)	1	0.1 (12.5%)	3	**0.2 (8.3%)**	**4**
Abandoned Fishponds	0.7 (41.2%)	1	-	-	**0.7 (29.2%)**	**1**
Total	**1.7 (100%)**	**12**	**0.8 (100%)**	**16**	**2.4 (100%)**	**28**

(1) Areas of ponds within Project Boundary that overlap with site areas of concurrent project, i.e. Development at Ngau Tam Mei Area, are excluded in the calculation as they would be lost under the concurrent project.

(2) Loss of whole pond has been assumed for partial encroachment of Project Boundary onto pond.

(3) The size and percentage included in this table have been subject to rounding adjustments. Any discrepancies between total and sums of individual numbers listed therein are due to rounding.

Loss of Active Fishponds

9.6.2.2 Within the Project Boundary, active fishponds under active production were mainly located in NTM area. A total of 25 active fishponds were identified in NTM area and three were located in Ki Lun Tsuen area (**Figure 9.4** and **Figure 9.6** refer). To minimise the direct impact to active fishponds, five fishponds (i.e. P#170-173 and P#244) are excluded from land resumption and no direct impact is expected. Quite a number of active fishponds in NTM area were utilised for cultivation of non-edible ornamental fish, such as Koi, while some were found to be utilised for cultivation of edible fish species.

9.6.2.3 Direct loss of fisheries production, resources and aquaculture activities at active fishponds (about 1.5 ha) (i.e. 23 fishponds) would be resulted from the Project (**Table 9.5** refers). The expected direct loss account for about 62.5% of the total fishpond areas within the Project Boundary. The expected direct loss also accounts for 0.1% of fishpond areas in Hong Kong in terms of aquaculture. Considering the area of direct loss of active fishponds does not account for a significant proportion of the total fishpond areas in Hong Kong and quite a number of active fishponds were utilised for cultivation of non-edible ornamental fish, the impact on fisheries resources and potential contribution to the annual fisheries production in Hong Kong is considered to be low.

Loss of Inactive Fishponds

9.6.2.4 For inactive fishponds, the Project would result in direct loss of 0.2 ha fishponds (i.e. four fishponds) in NTM area (**Figure 9.4** and **Table 9.5** refer). The expected direct loss of inactive fishponds accounts for about 8.3% of the total fishpond areas within the Project Boundary. In terms of fishpond areas in Hong Kong, the expected direct loss of inactive fishponds would account for <0.1%.

9.6.2.5 These inactive fishponds were observed without aquaculture activities during recent site visits. Given the small loss of inactive fishponds which does not account for a significant proportion of the total fishponds area in Hong Kong and no evidence of commercial aquaculture activities was observed in these inactive fishponds during recent observation, the impact on fisheries resources and fisheries production is considered very low.

Loss of Abandoned Fishponds

9.6.2.6 For abandoned fishponds, the Project would result in direct loss of 0.7 ha fishponds (i.e. one fishpond) in NTM area (**Figure 9.4** and **Table 9.5** refer). The expected direct loss of abandoned fishponds would be about 29.2% of the total fishpond areas within the Project Boundary. In terms of fishpond areas in Hong Kong, the expected direct loss of abandoned fishponds would account for about 0.1%.

9.6.2.7 The conditions of these abandoned fishponds are not suitable for short-term resumption of fisheries activities given the physical constraints such as inaccessibility, deterioration of pond bund, etc., and therefore not anticipated to support significant fisheries production. Furthermore, considering the small loss of abandoned fishponds does not account for a significant proportion of the total fishponds area in Hong Kong and no evidence of commercial aquaculture activities was observed in these abandoned fishponds during recent observation, the impact on fisheries resources and fisheries production is considered negligible.

Loss of Watercourses for Aquaculture Activities

9.6.2.8 Under the Project, some of the watercourses would be temporarily affected during the construction phase. However, except the small areas of pile cap areas (and associated viaduct piers) would be located within KTMDC, no further direct loss of watercourses/streams is anticipated under the Project, particularly for watercourses near NTMWTW. Moreover, the major water source for fishponds within the Assessment Area would be from direct rainfall. Thus, impact to aquaculture activities in terms of loss of water sources is not anticipated.

9.6.3 Construction Phase – Indirect Impact

Deterioration of Water Quality

9.6.3.1 Despite some existing fishponds were also observed directly outside the Project Boundary in NTM area, these fishponds are expected to be lost under concurrent project, thus, indirect impact to fishponds within concurrent project area is not anticipated. On the other hand, considering the proximity of the remaining active and inactive fishponds in the vicinity of the Project Boundary, e.g. the retained active fishponds to the west of NTMWTW and near Kei Lun Tsuen area, fishponds in Tung Shing Lei and to the southeast of NSW, indirect impact such as water quality impacts may arise from uncontrolled construction site run-off and accidental spillage, etc. during the construction phase.

9.6.3.2 Uncontrolled site run-off, sediment and chemical waste generated from construction activities could potentially cause water quality impacts, including parameters such as pH value, dissolved oxygen (DO) and salinity, to the adjacent watercourses and water bodies. Moreover, the use and storage of chemicals (e.g. engine oil and lubricants) during construction phase, may affect water quality if spillage occurs and enters the adjacent water bodies and watercourses. Groundwater pollution may also arise from improper handling and storage of chemicals within the works sites, where groundwater infiltrates. Without mitigation measure, these potential water quality impacts may result in potentially low to moderate impacts on the active and inactive fishponds in the vicinity of the Project Boundary.

9.6.3.3 The examples of relevant water quality control measures and the details of these mitigation measures are detailed in **Section 5**. Given the major water source for fishponds within the Assessment Area would be from direct rainfall, with proper implementation of the proposed mitigation measures, adverse impacts on water quality are not anticipated. As a result, no adverse impacts on aquaculture activities and fisheries production due to potential water quality deterioration is expected during the construction phase.

Water Table Drawdown, Disruption or Disturbance of Pond Culture Related Activities

9.6.3.4 Underground tunnelling works for tunnel section of the Project may potentially led to groundwater drawdown and affect the fishponds in the vicinity. Nonetheless, the tunnel sections of the Project would be in Ngau Tam Shan and to the east of Hadden Hill, away from existing fishponds (**Figures 9.4** to **9.6** and **Figure 9.8**). Besides, direct rainfall has been the major water source for fishponds within the Assessment Area. Thus, no adverse impacts on cultured fish due to potential water table drawdown is anticipated during the construction phase.

Potential Vibration Impact

9.6.3.5 Potential vibration impact may arise from the tunnelling works of the Project. Considering the underground works would be carried out in general at least 10 m below the local ground level and the tunnel sections of the Project would be in Ngau Tam Shan and to the east of Hadden Hill, i.e. away from the existing fishponds (**Figures 9.4** to **9.6** and **Figure 9.8**), potential vibration impact to fishponds is not anticipated.

Bund Stability

9.6.3.6 Construction activities that occur adjacent or in close vicinity to the fishponds located to the west of NTMWTW and near Ki Lun Tsuen area may affect the integrity of the pond bunds, potentially resulting in instability of the pond structure, or water seepage.

9.6.3.7 In the absence of mitigation measures, this indirect impact could inhibit pond usage and result in potentially low to moderate impact on the retained active fishponds to the west of NTMWTW and near Ki Lun Tsuen area. With appropriate mitigation measures implemented to ensure the stability of the pond bund during construction phase (e.g. sheet pile wall and associated grouting, further described in **Section 9.8.2.2**), adverse impacts on aquaculture activities are not anticipated.

Blockage of Access

9.6.3.8 During construction phase, construction activities may affect the existing access to the retained active fishponds to the west of NTMWTW and near Ki Lun Tsuen area. Without mitigation measures, this may result in low to moderate impact to these fishpond areas. Mitigation measures (e.g. provision of alternative access road and access arrangement) should be in place to minimise the potential blockage and the associated impact on fishpond areas during construction phase.

9.6.4 Operational Phase – Direct Impact

9.6.4.1 Based on the proposed Project layout, no additional direct impacts on fisheries resources (e.g. loss of fishponds) and associated fisheries production are anticipated during the operational phase of the Project. As described in **Section 9.6.2.1**, despite the proposed alignment would not be encroached into the fishponds to be impacted by temporary works, these fishponds (including 0.7 ha of active fishpond and 0.1 ha of inactive fishpond) are proposed to be included in the proposed wetland compensation area as an ecological enhancement measure of the overall NMH Project with expected benefits such as consolidating small and fragmented ponds into larger wetland with enhanced connectivity and physical modification of pond structures and pond bunds to allow more gently sloping banks for wildlife usage etc. Full-scale implementation of the wetland compensation could only proceed upon completion of permanent works and removal of temporary works. After the establishment period, the proposed wetland compensation site will be handed over to AFCD for habitat management, subject to prior written consent of AFCD and sufficient recurrent cost allocated to AFCD. The feasibility to include aquaculture activities in the proposed wetland compensation site would be explored after the establishment period of the proposed wetland compensation site.

9.6.5 Operational Phase – Indirect Impact

9.6.5.1 Based on the proposed Project layout and the nature of the Project, considered the permanent road drainage system of the Project would be properly designed and implemented, no indirect impacts on fisheries resources are anticipated during the operational phase of the Project.

9.6.6 Summary of Fisheries Impact

9.6.6.1 Based on the discussion above, the potential impacts on fisheries resources arising from the construction and operation of the Project, where present, are summarised in **Table 9.6** to **Table 9.8** below.

Table 9.6 Summary of Impact on Fisheries Resources (Active Fishponds) within Assessment Area

Criteria	Construction Phase	Operational Phase
Nature of impact	Direct impact: - Direct loss of active fishponds, quite a number of fishponds in NTM area were utilised for non-edible ornamental fish cultivation Indirect impact: - Reversible indirect water quality impacts to fishponds in the vicinity of the Project Site, due to potential construction	Direct impact: - Nil Indirect impact: - Nil

Criteria	Construction Phase	Operational Phase
	site run-off, accidental spillage, potential contamination of surface water and groundwater, bund stability and blockage of access.	
Size of affected area	Direct impact: - Direct loss of active fishpond: 1.5 ha Indirect impact: - Small area of active fishponds in the vicinity of the Project Site	Direct impact: - Nil Indirect impact: - Nil
Loss of fisheries resources / production	Direct loss of aquaculture resources and fisheries production within Project Boundary (quite a number of fishponds in NTM were utilised for non-edible ornamental fish cultivation)	Nil
Destruction and disturbance of nursery and spawning grounds	Not applicable (Remark: No nursery and spawning ground within assessment area)	
Impacts on fishing activity	Not applicable (Remark: No fishing activities within assessment area)	
Impact on aquaculture activity	Low to moderate direct impact and indirect impact in the absence of mitigation measures, considering quite a number of fishponds in NTM area were utilised for non-edible ornamental fish cultivation	Nil
Overall impact before mitigation	**Low**	**Negligible**

Table 9.7 Summary of Impact on Fisheries Resources (Inactive Fishponds) within Assessment Area

Criteria	Construction Phase	Operational Phase
Nature of impact	Direct impact: - Direct loss of inactive fishponds Indirect impact: - Reversible indirect water quality impacts to inactive fishponds in the vicinity of the Project Site, due to potential construction site run-off, accidental spillage and potential contamination of	Direct impact: - Nil Indirect impact: - Nil

Criteria	Construction Phase	Operational Phase
	surface water and groundwater	
Size of affected area	Direct impact: - Direct loss of inactive fishpond: 0.2 ha Indirect impact: - Inactive fishponds in the vicinity of the Project Site	Direct impact: - Nil Indirect impact: - Nil
Loss of fisheries resources / production	Considering no active aquaculture resources observed, no loss of fisheries production anticipated	Nil
Destruction and disturbance of nursery and spawning grounds	Not applicable (Remark: No nursery and spawning ground within assessment area)	
Impacts on fishing activity	Not applicable (Remark: No fishing activities within assessment area)	
Impact on aquaculture activity	No impact is anticipated on aquaculture activity	
Overall impact before mitigation	**Low**	**Negligible**

Table 9.8 Summary of Impact on Fisheries Resources (Abandoned Fishponds) within Assessment Area

Criteria	Construction Phase	Operational Phase
Nature of impact	Direct impact: - Direct loss of abandoned fishponds Indirect impact: - Nil	No adverse direct and indirect impacts are anticipated
Size of affected area	Direct impact: - Direct loss of abandoned fishpond: 0.7 ha Indirect impact: - Nil	Not Applicable
Loss of fisheries resources / production	Considering no active aquaculture resources observed, no loss of fisheries production anticipated	Not Applicable

Criteria	Construction Phase	Operational Phase
Destruction and disturbance of nursery and spawning grounds	Not applicable (Remark: No nursery and spawning ground within assessment Area)	
Impacts on fishing activity	Not applicable (Remark: No fishing activities within assessment Area)	
Impact on aquaculture activity	No impact is anticipated on aquaculture activity	
Overall impact before mitigation	**Negligible**	**Negligible**

9.7 Cumulative Impact

9.7.1.1 Cumulative impact may arise from the construction and operation of the Project, due to interaction with other developments in the area. The list of concurrent projects is stated in **Table 2.3** of **Section 2**. Major concurrent projects that may impose cumulative impact on aquaculture activities and production include the Northern Metropolis Highway – Kwu Tung Section, Northern Metropolis Highway – Tin Shui Wai Section, Development at NTM Area, Northern Link (NOL) Main Line, NTMWTW Extension and STLMC DN.

9.7.1.2 Given the assessment of Northern Metropolis Highway – Kwu Tung Section and Tin Shui Wai Section are still undergoing, relevant cumulative impact will be assessed in subsequent EIA reports. In terms of the loss of fishponds under the other concurrent projects within the Assessment Area, the loss under NOL Main Line would be short-term loss and would be released upon the completion of construction, while no direct loss of active fishponds is expected under NTMWTW Extension (MTRCL, 2023; WSD, 2024). On the other hand, fishponds loss under San Tin Technopole within the Assessment Area would be inactive and abandoned ponds to the northeast of San Tin Barracks (**Figure 9.5** refers). A suite of mitigation measures was proposed to enhance the fisheries resources (e.g. fisheries activities and production, culture area and aquaculture potential etc.) of the proposed Sam Po Shue Wetland Conservation Park with a view to compensate for the loss of fishponds arising from the development of the San Tin Technopole (CEDD, 2024). For Development at NTM area, the area of fishponds involved is considered as small in terms of the overall fishpond areas in Hong Kong and quite a number of the active fishponds were utilised for non-edible ornamental fish cultivation (CEDD and PlanD, 2025). As a result, the cumulative impact on fisheries is therefore considered as low.

9.8 Mitigation Measures

9.8.1.1 According to the general policy of mitigating fisheries impacts as discussed in Annex 17 of EIAO-TM, mitigation measures for minimisation of impacts to the affected fishponds are recommended below.

9.8.2 Minimisation

Minimisation of Potential Water Quality Impacts

9.8.2.1 In order to minimise the potential indirect fisheries impacts due to deterioration of water quality, including parameters such as pH value, dissolved oxygen (DO) and salinity, on the adjacent fishponds, mitigation measures and good site practices would be implemented during the construction phase, details refer to **Section 5**. Examples of guidelines for control of construction site run-off and construction-related activities, as well as other good site practices, are described below. With proper implementation of the proposed mitigation measures, adverse impacts on water quality are not anticipated.

Control of Construction Site Run-off

- Implementation of Best Management Practices, following the guidelines for handling and disposal of construction site discharges detailed in ProPECC PN 2/24 *Construction Site Drainage* and ETWB TC (Works) No. 5/2005 *Protection of Natural Streams / Rivers from Adverse Impacts Arising from Construction Works*;
- Controlling surface run-off from construction site via adequately designed channels, earth bunds or sandbag barriers, and directing the run-off to sand / silt removal facilities such as sand traps, silt traps and sedimentation basins before discharging to storm drains;
- Minimising soil excavation in wet season (April to September) as far as practicable, and proper covering of temporarily exposed slope surfaces, while intercepting channels should be provided along the crest / edge of excavation; and
- Proper covering of open stockpiles of construction materials during rainstorms (e.g. with tarpaulin or similar fabric).

Control of Construction-related Activities

- All vehicles and plants should be cleaned before they leave the construction site to minimise the deposition of earth, mud and debris in surrounding areas;
- Acidic wastewater generated from acid cleaning, etching, pickling and similar activities should be neutralised to within the pH range of 6 to 10 before discharging into foul sewers. If there is no public foul sewer in the vicinity, the neutralised wastewater should be transported off site for disposal into foul sewers or treated to a standard acceptable to storm drains and the receiving waters; and
- The Waste Disposal Ordinance (Cap. 354) and its subsidiary regulations in particular the Waste Disposal (Chemical Waste) (General) Regulation, should be observed and complied with for control of chemical wastes. The Contractor is also recommended to develop management procedures for chemicals used and prepare an emergency spillage handling procedure to deal with chemical spillage in case of accidents.

Maintaining Bund Stability

9.8.2.2 During the construction stage, retained active fishponds to the west of NTMWTW and near Ki Lun Tsuen area should be isolated and disconnected from any existing watercourses in order to minimise the potential water quality impact to the adjacent water bodies. A layer of shoring or sheet pile wall should be erected along the retained fishponds in order to maintain bund stability, where appropriate. In

addition, the shoring / sheet pile wall should have grouting or a grout curtain to avoid water seepage from the retained fishpond to the adjacent area. With the implementation of shoring / sheet pile, the stability of the retained fishpond bund should be preserved and the impacts on aquaculture activities should be minimised.

Other Good Site Practices

- Nomination of approved personnel, such as a site manager, to be responsible for implementation of good site practices, arrangements for waste collection and effective disposal to an appropriate facility;
- Training of site personnel in site cleanliness, concepts of waste reduction, reuse and recycling, proper waste management and chemical waste handling procedures;
- Provision of sufficient waste reception / disposal points, and regular collection of waste;
- Adoption of appropriate measures to minimise windblown litter and dust during transportation of waste by either covering trucks or by transporting wastes in enclosed containers;
- Provision of regular cleaning and maintenance programme for drainage systems, sumps and oil interceptors;
- Adoption of a recording system for the amount of wastes generated, recycled and disposed (including the disposal sites); and
- Preparation of Waste Management Plan (WMP), as part of the Environmental Management Plan (EMP).

9.9 Evaluation of Residual Fisheries Impacts

9.9.1.1 As described in **Section 9.6**, the overall fisheries impact would range from negligible to low. With the implementation of the proposed mitigation measures and good site practices listed in **Section 9.8**, adverse impacts on fisheries are not expected to arise. Hence, residual impact is not anticipated.

9.10 Environmental Monitoring and Audit

9.10.1.1 With the implementation of mitigation and the water pollution control measures proposed in **Section 5**, potential water quality impacts arising from the Project would be minimised. No specific EM&A programme is required for the potential water quality impact in association with fisheries impact. The monitoring and audit requirement have been covered by the EM&A programme for potential water quality impact recommended in **Section 5**.

9.11 Conclusion

9.11.1.1 Fisheries impact assessment based on the existing available information and site visits on pond culture fisheries resources and activities within the Assessment Area have been undertaken, following the criteria and guidelines as stated in Annexes 9 and 17 of the EIAO-TM, and the requirements in the EIA Study Brief (ESB-373/2025).

9.11.1.2 Under the Project, loss of fish culture area (i.e. fishponds) within the Project Boundary, including active fishponds (1.5 ha), inactive fishponds (0.2 ha) and

abandoned fishponds 0.7 ha), identified in NTM area and Ki Lun Tsuen within the Project Boundary is anticipated. To minimise direct impact to fishponds, five active fishponds (i.e. P#170-173 and P#244) are excluded from land resumption and would not be directly impact by the Project. Considering the fishpond area involved is small in terms of the overall fishpond areas in Hong Kong (i.e. less than 1% in total) and quite a number of active fishponds were utilised for cultivation of non-edible ornamental fish while no evidence of commercial aquaculture activities was observed in inactive and abandoned fishponds, the impact on fisheries resources and potential contribution to the annual fisheries production in Hong Kong arising from the Project is therefore anticipated to be low.

9.11.1.3 Upon the implementation of recommended mitigation measures, no adverse impact on fisheries resources and production is anticipated from the construction and operational phase of the Project.

9.12 References

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